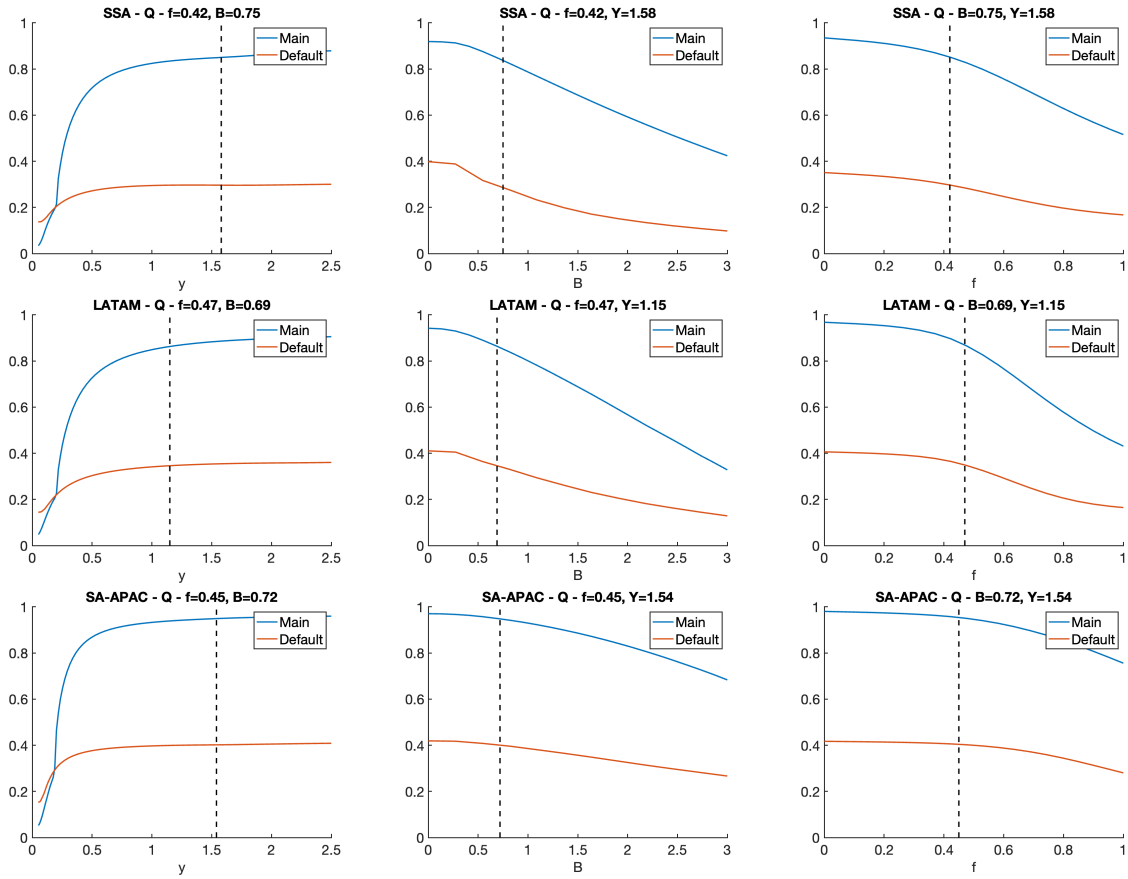

Figure 1: Q as a function of y (first column), B (second column) and f (third column): SSA (first line), LATAM (second line) and SA-APAC (third line). On top of each picture we report the other two state variables centered on median values. The dotted line represents the median value.

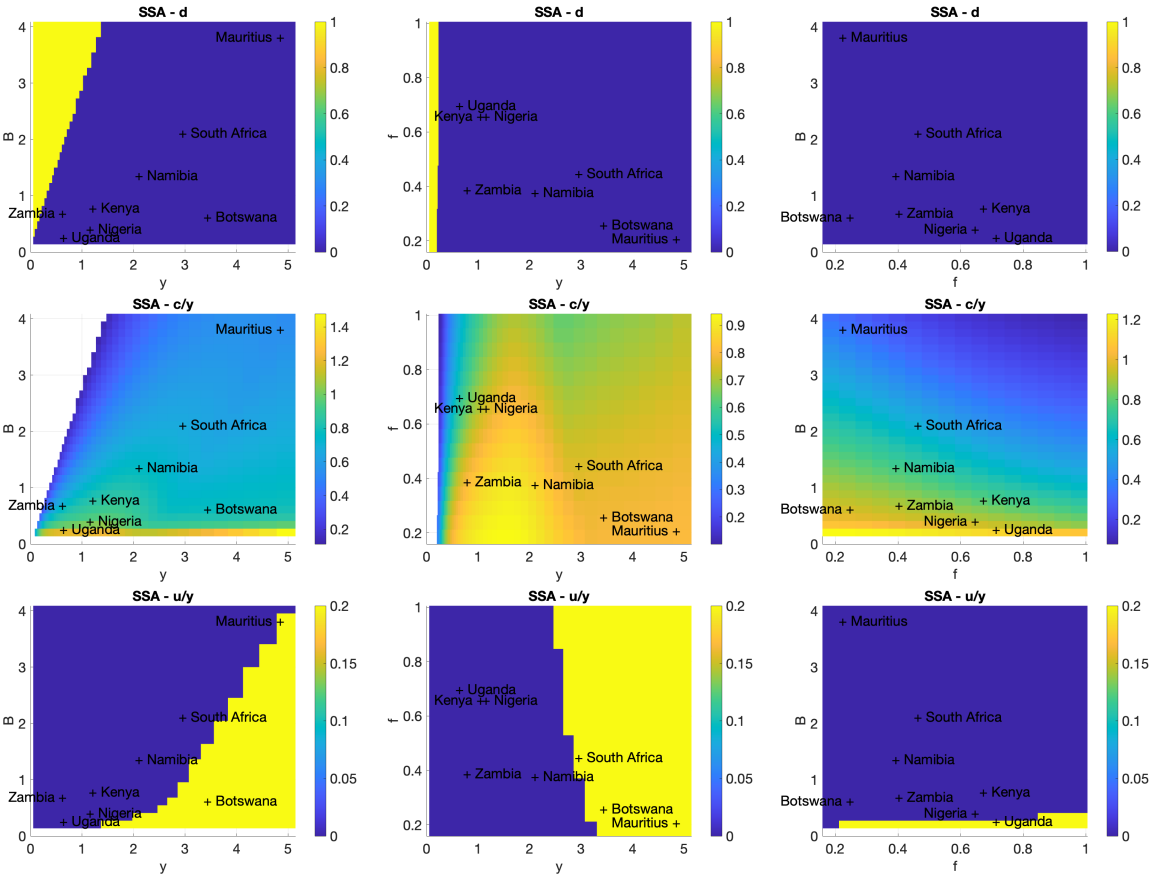

Figure 2: SSA region: decision to default (d), consumption over output (c/y) adaptation expenditure over output (u/y). When a variable is fixed, it is set equal to the median value for the macro-region, see Table 5: plane (y, B), $f = 0.42$; plane (y, f), $B = 0.75$; plane (f, B), $y = 1.58$. The white region in the picture for c/y corresponds to the default region. The positioning of the country labels in the pictures is purely indicative because, for example, in the (y, B) plane, the values of d, c/y, and u/y correspond to the median value of f (specifically, $f = 0.42$).

debt (3.36, 0.63) and the median value for f, i.e., 0.42, which is different from its own value (0.26).

In all the macro-regions, the decision to default only occurs for a very high debt to GDP ratio (around 250% for the SSA region, and 300% for SA-APAC and LATAM macro-region). On the output-pollution plane we observe that for a very low endowment the countries may opt to default, the default region enlarges as pollution increases but the effect is very limited. Climate risk doesn't lead to default on sovereign debt, in the pollution-debt plane, there is no space for the decision to default on debt. Notice that according to our model, no country should opt for default as shown in Tables 5-7.

As the stock of pollution increases, consumption over output decreases in all the three

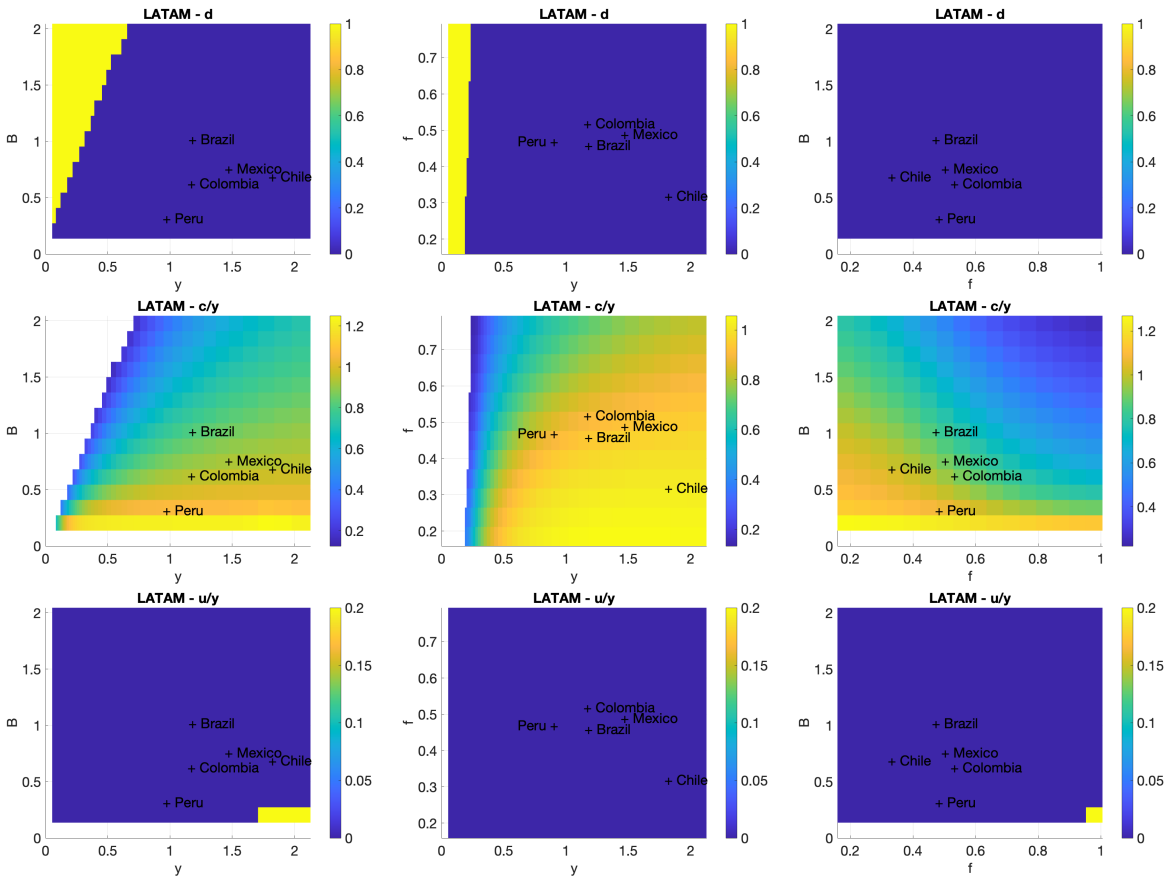

Figure 3: LATAM region: decision to default (d), consumption over output (c/y) adaptation expenditure over output (u/y). When a variable is fixed, it is set equal to the median value for the macro-region, see Table 6: plane (y, B), $f = 0.47$; plane (y, f), $B = 0.69$; plane (f, B), $y = 1.15$. The white region in the picture for c/y corresponds to the default region. The positioning of the country labels in the pictures is purely indicative because, for example, in the (y, B) plane, the values of d, c/y, and u/y correspond to the median value of f (specifically, $f = 0.47$).